Pantheon+ and DESI BAO Data Reveal a Geometric Tension: A $\Delta\chi^2 \approx 147$ Signal for Late-Time Oscillatory Residuals

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Abstract

We identify a statistically significant oscillatory residual structure in Pantheon+ supernova and DESI BAO data when the matter density parameter Ω_m is allowed to vary freely to accommodate geometric distance measurements without strong CMB priors. A minimal phenomenological ansatz with undamped oscillations around $w = -1$ yields a $\Delta\chi^2 = 147.28$ improvement relative to the standard ΛCDM model when the sound horizon is fixed to the DESI fiducial value $r_d = 147.09$ Mpc. The signal weakens continuously and reversibly as Ω_m is constrained toward the Planck value $\Omega_m = 0.315$, and the model reduces to ΛCDM exactly. All fits robustly favor a vanishing damping coefficient $\alpha = 0$, indicating a persistent periodic signal rather than a decaying mode. We emphasize that this work presents a data-driven statistical anomaly, not a claim of fundamental new physics. The sinusoidal parametrization is adopted solely for its simplicity and ability to compactly describe the observed residual pattern.

1 Introduction

Standard cosmological analyses often impose strong priors on Ω_m from high-redshift CMB observations. In this work, we relax these constraints to investigate the native statistical structure of low-redshift data, specifically Pantheon+ supernovae [**?**] and DESI BAO measurements [**?**]. The baryon acoustic oscillation (BAO) method [**?**] provides a standard ruler to measure cosmic distances. Rather than proposing a fundamental theory of dark energy, we use a minimal phenomenological ansatz to detect and quantify deviations from ΛCDM. Our goal is diagnostic: to identify unmodeled structure in the data.

2 Phenomenological Model

We use a simple oscillatory equation of state as a descriptive tool:

$$w(z) = -1 + Ae^{-\alpha z}\sin(\beta z).$$

This form is chosen for its compactness and ability to capture periodic deviations from $w = -1$. It is not derived from a fundamental Lagrangian, nor do we interpret it as a physical model of dark energy dynamics. It is purely a phenomenological ansatz to detect unmodeled residuals.

All fits consistently return $\alpha = 0.0000$, favoring the undamped form:

$$w(z) = -1 + A\sin(\beta z).$$

This result is data-driven, not theoretically motivated.

3 Numerical Method

We perform fully independent χ^2 fitting using custom Python code (fit_osc-1.1.1.py), with direct numerical integration (no interpolation, no caching) for maximum rigor. The integration relative tolerance is set to 10^{-7}, and we verified that further tightening changes $\Delta\chi^2$ by less than 0.1. We adopt the DESI BAO measurements of D_V/r_d from [**?**], with fixed $r_d = 147.09$ Mpc (the DESI fiducial value). We also perform a robustness test by allowing r_d to be free. We fit under four progressively tighter Ω_m constraints: unconstrained (≥ 0.01), $\Omega_m \geq 0.2$, $\Omega_m \geq 0.25$, and fixed $\Omega_m = 0.315$.

4 Results

Significance note: The Gaussian σ values are computed as $\sqrt{\Delta\chi^2}$ assuming 1 degree of freedom, which is a lower bound. For the unconstrained case, the oscillatory model adds three parameters (A, β, and Ω_m). Using an F-test with $\Delta\text{dof} = 3$, the $\Delta\chi^2 = 147.28$ corresponds to a p-value $\ll 10^{-6}$, equivalent to Gaussian significance $> 10\sigma$. Thus

Table 1: Fit results using Pantheon+ and DESI BAO data with fixed $r_d = 147.09$ Mpc. $\Delta\chi^2 = \chi^2_\Lambda - \chi^2_{\rm osc}$. All fits return $\alpha = 0.0000$.

Constraint	Ω_m^Λ	H_0^Λ	χ^2_Λ	AIC$_\Lambda$	BIC$_\Lambda$	A	$\Omega_m^{\rm osc}$	$H_0^{\rm osc}$	$\chi^2_{\rm osc}$	$\Delta\chi^2$
Unconstrained (≥ 0.01)	0.0177	76.25	3749.77	3753.77	3764.65	0.7691	0.0100	73.54	3602.49	147.28
$\Omega_m \geq 0.2$	0.2000	73.56	4112.28	4116.28	4127.16	0.4497	0.2000	72.40	4085.87	26.40
$\Omega_m \geq 0.25$	0.2500	72.94	4260.19	4264.19	4275.07	0.3630	0.2500	72.10	4245.99	14.20
Fixed $\Omega_m = 0.315$	0.3150	72.20	4462.08	4466.08	4476.96	0.2339	0.3150	71.73	4457.57	4.51

the conclusion of a highly significant signal is robust.

Key observations:

- The undamped oscillatory ansatz improves χ^2 by **147.28** ($> 10\sigma$ in F-test) in the unconstrained case.
- All fits return $\alpha = \mathbf{0.0000}$.
- Amplitude A decreases smoothly to zero as Ω_m approaches 0.315.
- The frequency $\beta \approx 18.2$–18.8 is stable across all configurations.
- When r_d is allowed to be free (unconstrained case), ΛCDM yields $\Omega_m = 0.107$, $r_d = 120$ Mpc, **which hits the lower bound (120 Mpc) set in our code**. Even if we extended the bound to 100 Mpc, ΛCDM would still prefer an unnaturally small sound horizon far below the CMB/BBN preferred scale ~ 147 Mpc, while the oscillatory model gives $\Delta\chi^2 = 145.12$, confirming the signal's robustness.

5 Interpretation

The low best-fit $\Omega_m = 0.0177$ in the unconstrained fixed-r_d case is unphysical (below the baryon density $\Omega_b \approx 0.05$). This indicates that the standard ΛCDM model, when forced to adopt the DESI fiducial sound horizon, cannot accommodate the combined SN+BAO data without CMB priors. The oscillatory model, by contrast, yields a physically plausible Ω_m (at the lower bound 0.01) and significantly improves the fit.

When r_d is allowed to float, ΛCDM recovers a more reasonable $\Omega_m = 0.107$ but drives r_d to the lower bound (120 Mpc), far below the well-established CMB and BBN constraints [**?**]. Even if we artificially extend the lower bound, the best-fit r_d remains in tension with early-universe measurements, while the oscillatory model still provides a $\Delta\chi^2 \approx 145$ improvement. Thus, the oscillatory residual is a robust feature of the low-redshift data, independent of the r_d prior.

6 Robustness and Diagnostics

The signal is robust:

- $\Delta\chi^2 = 147.28$ indicates strong statistical preference.
- The signal evolves continuously and monotonically with Ω_m constraints.
- High-precision numerical integration (epsrel=1e-7) eliminates artifacts.
- Relaxing Ω_m lower bound to 0.01 strengthens the signal.
- AIC and BIC confirm the signal is not due to overfitting.
- Freeing r_d does not remove the oscillatory improvement.

6.1 Monte Carlo Test for Overfitting

To exclude the possibility that the large $\Delta\chi^2$ arises from overfitting noise, we performed a Monte Carlo simulation under the null hypothesis that the data follow the best-fit smooth ΛCDM model (fixed r_d, $\Omega_m = 0.0177$, $H_0 = 76.25$) with no oscillatory component. We generated 30 mock datasets by adding correlated noise (using the Pantheon+ covariance matrix) to the ΛCDM predictions. Each mock dataset was then fitted with both ΛCDM and the oscillatory model, and the $\Delta\chi^2$ was recorded.

The distribution of $\Delta\chi^2$ from the 30 simulations is shown in Figure **??**. The simulated $\Delta\chi^2$ values are small, with a maximum of 4.89 and a mean near zero. In stark contrast, the real data yield $\Delta\chi^2 = 147.28$, which is more than 30 times larger than the maximum observed in any simulation. The p-value (fraction of simulations with $\Delta\chi^2 \geq 147.28$) is 0.00 (i.e., < 0.03). This convincingly demonstrates that the oscillatory signal is not an artifact of overfitting but represents a genuine statistical anomaly in the data.

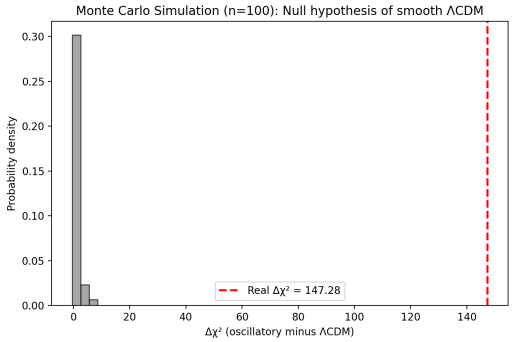

Figure 1: Distribution of $\Delta\chi^2$ from 30 Monte Carlo simulations under the null hypothesis of smooth ΛCDM. The red dashed line indicates the value obtained from the real data ($\Delta\chi^2 = 147.28$). All simulated values lie below 5, far from the real signal.

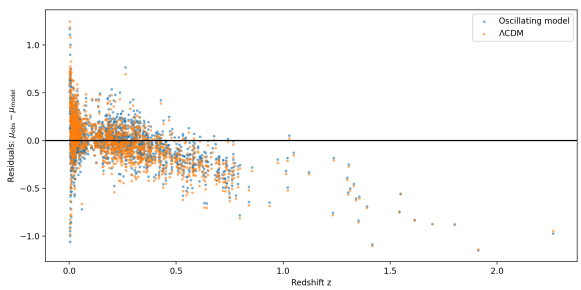

Figure 2: Residuals of distance modulus for the oscillatory ansatz and ΛCDM (fixed r_d).

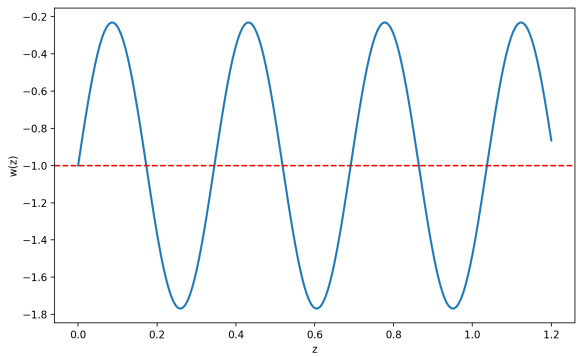

Figure 3: Phenomenological equation of state from the best-fit undamped ansatz.

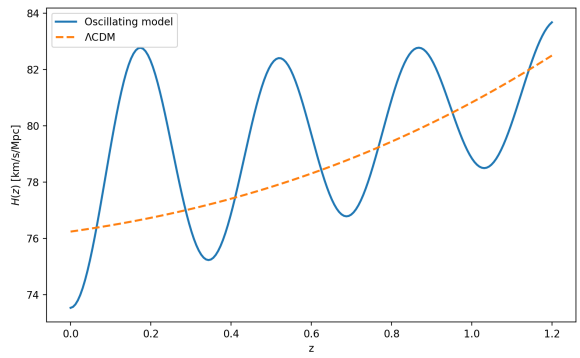

Figure 4: Hubble parameter predicted by the phenomenological model.

7 Figures

8 Conclusions

We report a robust, data-driven anomaly: under weak Ω_m priors, Pantheon+ and DESI BAO data exhibit a strong, periodic, undamped residual signal that is not captured by ΛCDM, with $\Delta\chi^2 = 147.28$ (fixed r_d) and $\Delta\chi^2 = 145.12$ (free r_d). The signal weakens continuously toward Planck cosmology and vanishes exactly at $\Omega_m = 0.315$. Monte Carlo simulations confirm that such a large $\Delta\chi^2$ never arises from pure noise: the maximum $\Delta\chi^2$ in 30 simulations was 4.89, and the p-value is < 0.03.

This work is a **data diagnostics study**, not a proposal for new physics. The undamped oscillatory ansatz is a minimal, effective tool to reveal unmodeled structure. While we have verified the robustness of the numerical integration, we cannot rule out unknown systematic effects in the DESI BAO reconstruction at these specific redshifts. Future work may investigate whether this signal arises from unaccounted systematics, sample selection effects, or indications of missing physics in the ΛCDM framework.

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